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import java.io.IOException;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class LogAnalysis {

    public static class LogMapper
        extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text logType = new Text();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String line = value.toString();

            if (line.contains("ERROR"))
                logType.set("ERROR");
            else if (line.contains("WARNING"))
                logType.set("WARNING");
            else
                logType.set("INFO");

            context.write(logType, one);
        }
    }

    public static class LogReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {

        public void reduce(Text key, Iterable<IntWritable> values,
            Context context)
            throws IOException, InterruptedException {

            int sum = 0;
            for (IntWritable val : values) {

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        sum += val.get();
    }
    context.write(key, new IntWritable(sum));
}
}

public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Log Analysis");

    job.setJarByClass(LogAnalysis.class);
    job.setMapperClass(LogMapper.class);
    job.setReducerClass(LogReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Input:

INFO User logged in
 ERROR Disk failure
 INFO File uploaded
 WARNING Memory low
 ERROR Network issue

```

Merged Map outputs=1
GC time elapsed (ms)=53
Total committed heap usage (bytes)=514850816
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=100
File Output Format Counters
Bytes Written=37
C:\logproject>type output1\part-r-00000
ERROR 2
INFO 2
WARNING 1

```